



AI for GitHub: A View from Next

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GitHub Next

Researching the future of
software development

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TLDR

The big game is using **LLMs** to bring **assists** to **dev activities**.

Many possible **assists**

Many different **activities**

Delivery in **editor**, **monolith**, **application** and **command line**

LLMs are **probabilistic text calculators**

Text in, text out.

Key topics: prompts, context, control, quality, divide-and-conquer, hallucination.

LLMs are **magical** but **difficult** functions to integrate in **high-trust scenarios**.

Scepticism is healthy. Almost nothing but search in github.com today is probabilistic.

The human is almost certainly in-the-loop.



Announced yesterday!

Copilot for Docs

March 22, 2023

WAITLIST

How would it feel to have an expert on hand at all times? We built a tool that uses relevant information from a project's documentation to answer questions or explain concepts.

Copilot for Pull Requests

March 22, 2023

WAITLIST

Pull requests are a central part of the GitHub user experience. Copilot for PRs brings the power of Copilot to the PR experience, to help you write better PR descriptions, and to help your team review and merge PRs faster.

Copilot for CLI

March 22, 2023

WAITLIST

Ever having trouble remembering that shell command or this obscure flag? Don't worry: we're building GitHub Copilot assistance right into your terminal

Copilot Voice

March 22, 2023

WAITLIST

Write code without the keyboard. Difficulty typing? Use your voice to code without spelling things out by talking with GitHub Copilot.



What is needed to contribute?

Understand the rationale and business case

Get some hands-on experience with LLMs, esp GPT-3/4

Get in a playground and try things out. Be sceptical, push the limits.

Put some code in as input and output. Invoke from APIs and build an understanding of latencies, costs, reliability.

Understand the design principles

I'll outline my view of some of these. There are other excellent resources. Develop your own point of view.

Get familiar with an end-to-end AI codebase for **service** and **UX**

Several available.

QA, telemetry, nature of workloads etc.

Learn about Responsible AI and other training topics.



Sample Repo (Copilot4PRs)

- Learn about it from the AI generated summaries!

github.com/githubnext/prbot

github.com/githubnext/prbot/blob/thumbs/main/.thumbs/thumb.md

- Topics of interest

- Prompts (e.g. see tests)
- Prompt-building for many items
- Hierarchical summarization
- Test generation
- Multi-modal: cmdline, bot, service, vscode
- Lots of utilities
- QA: telemetry, baselines, scoring, deltas, rapid iteration (local caching of requests)



AI Outputs Overview

- **Summarization**
- Indexing
- Chat
- Generative
- Critique



AI Outputs Overview

- Summarization

- Keywords

- Title

- Summary

- Abstract

- Walkthrough

- Documentation

- Indexing

- Chat

- Generative

- Critique



Summarization: Understand the Rationale

We are headed to a world where much **text associated with code/repos** is **emergent** and **automatic**.

Content ⇒ **Search**

Content ⇒ **Text**

LLM AI is now able to **summarize** and **explain** code and changes pretty well. Analogies:

- | | | |
|--|---------------------|------------------|
| ● Abstracts, keywords, summaries, descriptions | = for comprehension | |
| ● Table of contents | | = for navigation |
| ● Documentation, walkthroughs | = for explanation | |

Use cases:

- | | | |
|---|--------|-------------------|
| ● Tool-tips | | Infopanel content |
| ● Seeding human-authored descriptions | Search | |
| ● Most useful for those unfamiliar with the content, or pressed for time , or small-factor devices . | | |

Risks:

- Bad UX, Too Much Information, Distracting
- Inaccurate, misleading
- Too Much Text for non-native speakers



Summarization: Understand the Business Case

Possibilities for \$\$\$

- Direct use in UX
 - Descriptions available everywhere
 - Monolith
 - Editor (VSCode, ...)
 - Files, directories, changes, issues, PRs, projects, ...
- Supporting human
 - Completions
 - Editable descriptions and documentation
 - Explaining the swamp of legacy code - “Undocumented PR from dev 8yr ago”
- Feed into other tools



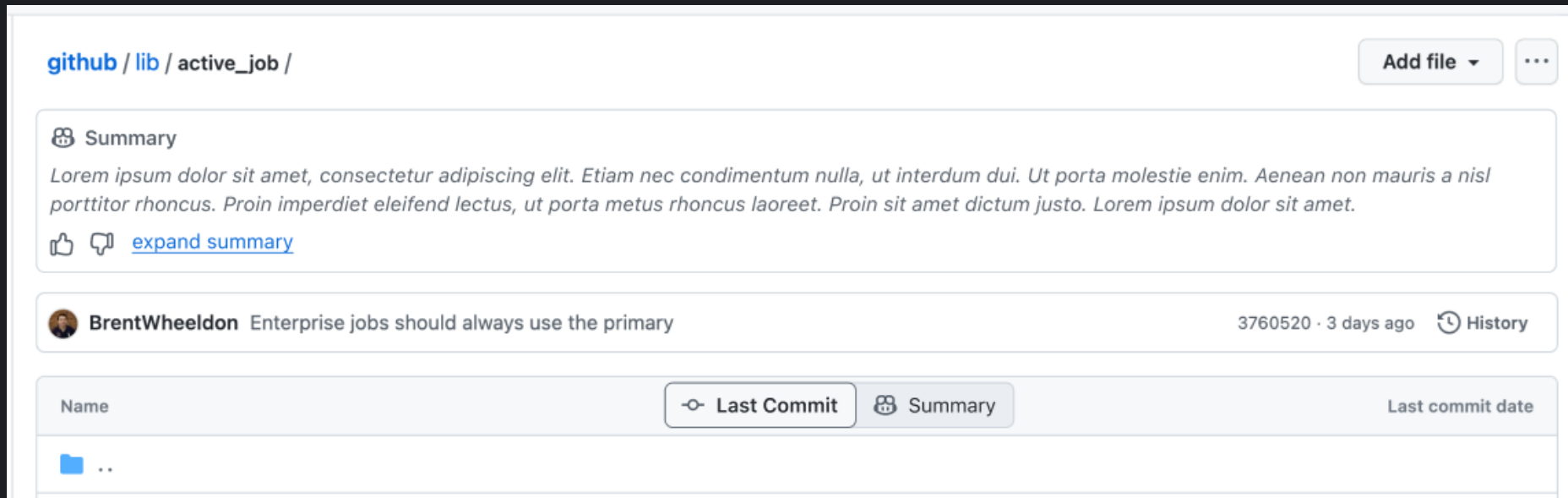
AI Summarization – Example – PR Summary

`./prbot describe https://github.com/githubnext/prbot/pull/565`

▼ Summary

This pull request enhances the bot's ability to describe code changes, updates the test scores for the describe task, and simplifies the poetry generation code. It modifies `src/bot/bot-describe.ts` to improve the description quality and readability, updates `test/scores/boosted/describe-score.md` to report the latest results, and removes an unused case from `src/engine/poems.ts` to improve code clarity.

AI Summarization – UX Mockups (EPD Core)



AI Summarization – UX Mockups (EPD Core)

github / lib / active_job /

Commit

Summary

Add file

 **BrentWheeldon** Enterprise jobs should always use the primary

3760520 · 3 days ago

History

Name	Summary
 ..	
 queue_adapters	Summary text here
 enqueue_failed_error.rb	Summary text here
 initially_enqueued_at.rb	Summary text here
 job_class_actor.rb	Summary text here, summary text here, summary text here, summary text here, summary te...
 locking_job.rb	Summary text here
 scheduled_job.rb	Summary text here
 tenant_context.rb	Summary text here

AI Summarization - Sample Short Content

 scorer	<i>Scripts and data for evaluating prbot summaries and walkthroughs.</i>	2 weeks ago
 src	<i>Source code for prbot, a GitHub bot using OpenAI models.</i>	3 hours ago
 test	<i>Tests, baselines, and scores for the prbot tool.</i>	4 hours ago
 .dockerignore	<i>Files and directories to exclude from Docker build context.</i>	6 months ago
 .env.example	<i>Environment variables for prbot, including credentials and proxy URL.</i>	4 months ago
 .eslintignore	<i>Files and directories to exclude from ESLint.</i>	4 months ago
 .eslintrc	<i>ESLint configuration for TypeScript files with plugins and rules.</i>	4 months ago

locking_job.rb

205 lines (205 loc) · 8.62 KB

File summary

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Etiam nec condimentum nulla, ut interdum dui. Ut porta molestie enim. Aenean non mauris a nisl porttitor rhoncus. Proin imperdiet eleifend lectus, ut porta metus rhoncus laoreet. Proin sit amet dictum justo. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia curae;



 [locking_job.rb](#)

 [scheduled_job.rb](#)

 [tenant_context.rb](#)

ast commit message

enterprise jobs should always

ump to typed true

et lib to typed: false

ump to typed true

Merge remote-tracking branch

set lib to typed: false

Adds Lint/UnusedBlockArgument

Some Principles of AI Features in GitHub

- Utility
- Quality
- Trust
- Handling Change
- User Promotion



Principles of AI Features

- **Utility**

- The AI outputs must make some human task faster, or better
- All new outputs take user attention. They aren't free.

- **Quality**

- **Trust**

- **Handling Change**

- **User Promotion**



Principles of AI Features

● Utility

● Quality



The AI makes mistakes, these must be minimized, but still happen

Longer outputs → More mistakes (non-linear?)

The user must either approve/reject or tolerate or not care

Users are more tolerant of mistakes in ephemeral outputs

● Trust

● Handling Change

● User Promotion



Principles of AI Features

- Utility

- Quality

- Trust



- Mistakes are highly corrosive to trust

- Authorship must be understood

- The user must understand that they are reading AI outputs

- A human can explicitly take ownership after checking (cf. TAB in Copilot)

- Handling Change

- User Promotion



Principles of AI Features

- Utility

- Quality

- Trust

- Handling Change

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Like search, AI outputs should be time-labelled, fresh and up-to-date
User-editing of outputs is often requested but conflicts with change

- User Promotion



Principles of AI Features

- Utility

- Quality

- Trust

- Handling Change

- User Promotion



- AI outputs labelled until explicitly promoted



- In editor: ghost suggestions, gesture (e.g. TAB) to accept



- Out of editor: thumbs-up/down, edit, approve





Thank you!

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